Request for Information - Arresting Cable Release Mechanism (ACRM)

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INTRODUCTION

The United States (U.S.) Army Contracting Command - Rock Island, Rock Island, IL 61299, is issuing this RFI for the development of an Arresting Cable Release Mechanism (ACRM) capability for the Mine Clearing Line Charge (MICLIC). This RFI is intended to conduct market research to identify potential sources interested in providing information on current and emerging technologies.

The Army is seeking innovative / mature components and / or integrated solutions to support the ACRM development effort.

The ACRM Integrated Product Team will review white paper responses to this RFI. This approach enables the Army to identify, analyze, and evaluate innovative, mature, and emerging technologies.

PURPOSE

The Army is seeking an innovative, mature, modular, and / or integrated capability to automatically sever / disconnect the arresting cable (3/4 inches nylon rope with two MIL-C-13486-1 electrical wires woven inside) while under tension (approximately 1,000 lbs).

Background

US Army's M1150 Assault Breacher Vehicle (ABV) utilizes the Mine Clearing Line Charge (MICLIC) ammo components, specifically the M58 Linear Demolition Charge (LDC) and MK22 mod 4 rocket. The Linear Demolition Charge (LDC) will be deployed by the launch of the MK22 mod 4 rocket. An arresting cable stops the rocket and provides a safe standoff distance. It takes approximately 7 seconds after rocket initiation for the deployed LDC and the MK22 mod 4 rocket to settle fully on the ground. The ABV operator will proceed to detonate the LDC. After the flying debris are settled, the current procedure requires manual severing / disconnecting the arresting cable to prevent entanglement. The ACRM concept should automate this process by severing / disconnecting the arresting cable under tension (after rocket launch but prior to the detonation of the LDC) thus throwing it clear of the vehicle. The arresting cable must be released while under tension.

Figure 1.

Figure 1. shows the arresting cable (under tension and no tension) tethered to the M58 LDC container.

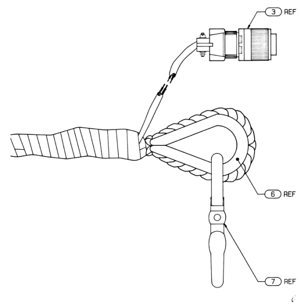

Figure 2.

Figure 2. shows an illustration of typical arresting cable and chain coupler or snap hook (REF 7).

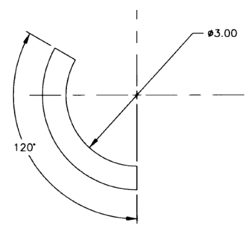

Figure 3.

Figure 3. shows the partial 0.5 inches steel ring for attaching the arresting cable.

Figure 4.

Figure 4. shows the anchor point for attaching the arresting cable.

Detailed interface drawings are available but are Distribution Statement D, distribution is authorized to Department of Defense (DoD) and U.S. DoD Contractors only. Interested contractors who would like to obtain any of these drawings must submit a written request to the POC identified below. The written request must come from the data custodian of the requesting company who is registered with the Joint Certification Program (JCP). The request must include the company's cage code and Joint Certification Number. More information on the JCP may be found at: http://www/dlis.dla.mil/jcp/. Interested contractors may be required to execute and return a Non-Disclosure and/or Non-Use Agreement prior to obtaining the drawings.

INFORMATION REQUESTED

Interested parties are requested to submit a white paper describing innovative, mature / modular components, and / or integrated capabilities that enable the severing /

disconnect of the arresting cable. The envisioned ACRM will consist of two major subsystems / functions.

1. Modular Triggering Mechanism / Methodology – can leverage any environmental conditions (command signal to launch the rocket, rocket launch, arresting of the rocket, etc.) prior to the detonation of the LDC.

    a. If a tension based triggering mechanism is to be placed between M58 LDC container connection point (see figure 1. and 3.) and the arresting cable, the triggering mechanism must survive the peak tension during the arresting event. The steady state tension on the arresting cable is approximately 1000 lbs.

2. Modular Release Mechanism – Any releasing method (electrical, mechanical, pyrotechnic, etc.) will be considered. However, the release must be completed within 3-45 sec after the settling of LDC and rocket on the ground (approximately 7 sec after rocket launch).

    a. Must be installed between the M58 LDC container connection point (see figure 1. and 3.) / chain coupler, the chain coupler / thimble or on the ¾" nylon arresting cable rope (Figure 2.). Replacement of the chain coupler by a proposed release mechanism is acceptable. Installation procedure must be executed at time of loading the M58 LDC container onto the ABV by Soldiers, not at the factory. Installation should occur with minimal disturbance to the packed line charge. Available free space between packed line charge and arresting cable anchor point is limited see figure 4.

    b. Must fully sever / disconnect arresting cable from the M58 LDC container.

    c. If cutting method is proposed, such as severing of the ¾" nylon rope, please be advised that there are two MIL-C-13486-1 electrical wires woven into the 3/4" nylon rope.

    d. Timing: Release must occur no earlier than 3 seconds after the LDC settles fully on the ground and not more than 45 seconds after the LDC settles fully on the ground. If leveraging rocket launch for triggering, LDC settles fully on the ground approximately 7 seconds after rocket initiation.

    e. Power source: Any method will be considered (electrical, pyrotechnic, mechanical, etc.). If electrical, there is strong preference for battery operated concepts to avoid additional cabling and maximize design portability from the ABV platform to emerging robotic platforms.

    f. Must not interfere with deployment of the LDC from the M58 LDC container.

g. Must survive severe MICLIC launch environment to include max/peak arrest force of 10,500 pounds.

Note: Capability that combines both triggering and release mechanisms in a single standalone unit is also acceptable.

RESPONSE INSTRUCTIONS

Please provide a determination as to whether the information contained in this white paper is subject to the International Traffic in Arms Regulations (ITAR), the Export Administration Regulations (EAR), or other export control regulations. If ITAR-controlled, please identify the relevant United States Munitions List category(ies). If EAR-controlled, please provide the Export Control Classification Number.

If the system cannot provide all the capabilities listed in this RFI, the white paper must clearly indicate which capabilities are being addressed. The white paper must include background on vendor capabilities and experience, a description of their technical approach to provide the identified capabilities. The white paper shall not be longer than 10 pages plus a cover page. Finally, the white paper must provide Technology Readiness Levels (TRL) for the proposed capabilities meeting the capabilities descripted under “information requested” section. Please see the Department of Defense’s (DoD) Technology Readiness Assessment Guidebook, Feb 2025.

The content and format for submission is outlined below. All submissions shall be in either Microsoft Word or searchable Portable Document Format. In addition to the title page, which must contain company name and submittal title at a minimum, submittal length must be no more than the following: Sections 1 and 2 up to eight pages, Sections 3 through 4 up to two pages.

1. Background (Company and technical background):

   a. Company overview

   b. Company size

   c. Relevant experience

   d. Key differentiators: What makes you different from other vendors?

2. Technical Approach

   Respondents are free to provide white paper on individual capability and / or an integrated solution described under the “INFORMATION REQUESTED” section. Regardless of approach, respondents should provide white paper and supporting material that include, but not be limited to the following:

   a. Detailed description

b. Theory of operations
c. Dependent technologies
d. Required enablers
e. Complexity to integrate proposed capabilities
f. Ease of use/installation
g. Reusability – this is not a requirement, but a reusable solution would be desirable
h. Response time
i. Reliability
j. Size Weight and Power (SWaP) characteristics
k. Examples of real-world application
l. Material availability and Lead time
m. Country of origin for critical components
n. Unit production costs. etc.

3. Assumptions and Risks

Please list any assumptions made in your response. Please list any significant risks to successful implementation of your concept.

4. Past Investment / Defense Contracts

a. Brief description of similar projects/clients; including strategic partnerships

b. Past performance for relevant capabilities (i.e. DoD / International Defense Contracts / other demonstrations or test events, prior work experience, joint ventures, etc.)

c. Contract Number (if applicable) and Service Component

(a) Brief description of project; include the project name if applicable and strategic partnerships

Respondents shall not be obligated to provide the services described herein and it is understood by the U.S. Government that the cost estimates provided because of this request are “best” estimates only.

Interested parties must submit a response no later than 30 days from the posting of this announcement. All responses to this RFI shall be submitted via email to danielle.d.miller3.civ@army.mil.